

1. Prob 16.4

(a) If we model the electron as a spherical shell of charge of radius a , the charge density is given by

$$\rho(\mathbf{x}) = \frac{e}{4\pi a^2} \delta(r - a) \quad (1)$$

then by (16.30), we have

$$M(\omega) = m_0 + \frac{2}{3c^2} \int d^3x \int d^3x' \rho(\mathbf{x}) \frac{e^{i\omega R/c}}{R} \rho(\mathbf{x}') = m_0 + \frac{2}{3c^2} \underbrace{\int d\Omega \int d\Omega' \left(\frac{e}{4\pi}\right)^2 \frac{e^{i\omega R/c}}{R}}_I \quad (2)$$

By (9.98),

$$\frac{e^{i\omega R/c}}{R} = 4\pi \left(\frac{i\omega}{c}\right) \sum_{l=0}^{\infty} j_l\left(\frac{\omega r_{<}}{c}\right) h_l^{(1)}\left(\frac{\omega r_{>}}{c}\right) \sum_{m=-l}^l Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) \quad (3)$$

and by addition theorem (3.62)

$$\sum_{m=-l}^l Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi) = \left(\frac{2l+1}{4\pi}\right) P_l(\cos \gamma) \quad (4)$$

With spherical symmetry in one of the two integrals, we can express I as

$$I = 4\pi \int \sin \gamma d\gamma d\phi \left(\frac{e}{4\pi}\right)^2 \cdot 4\pi \left(\frac{i\omega}{c}\right) \sum_{l=0}^{\infty} j_l\left(\frac{\omega a}{c}\right) h_l^{(1)}\left(\frac{\omega a}{c}\right) \left(\frac{2l+1}{4\pi}\right) P_l(\cos \gamma) \quad (5)$$

Notice that $\sin \gamma d\gamma = -d \cos \gamma = -P_0(\cos \gamma) d \cos \gamma$, only $l = 0$ in (5) survives due to the orthogonality of Legendre polynomial, thus

$$\begin{aligned} I &= 4\pi \cdot 2\pi \left(\frac{e}{4\pi}\right)^2 4\pi \left(\frac{i\omega}{c}\right) j_0\left(\frac{\omega a}{c}\right) h_0^{(1)}\left(\frac{\omega a}{c}\right) \frac{1}{4\pi} \int_0^\pi \sin \gamma d\gamma \quad \xi \equiv \frac{2\omega a}{c} \\ &= \frac{e^2}{a} \left(\frac{e^{i\xi} - 1}{i\xi}\right) \end{aligned} \quad (6)$$

giving

$$M(\omega) = m_0 + \overbrace{\frac{2e^2}{3ac^2}}^m + \frac{2e^2}{3ac^2} \left(\frac{e^{i\xi} - 1 - i\xi}{i\xi}\right) \quad (7)$$

(b) To the lowest order,

$$e^{i\xi} - 1 - i\xi = -\frac{\xi^2}{2} + O(\xi^3) \quad (8)$$

therefore

$$M(\omega) \approx m + \frac{2e^2}{3ac^2} \frac{i\xi}{2} = m + \frac{e^2}{3ac^2} \cdot \frac{2i\omega a}{c} = m(1 + i\omega\tau) \quad \tau = \frac{2e^2}{3mc^3} \quad (9)$$

The zero $\omega = i/\tau$ is in the upper half-plane.

Similar to the discussion (point 3) in section 16.3, physically, the Fourier-domain equation of motion (16.29) indicates that the velocity

$$\mathbf{v}(\omega) = \frac{\mathbf{F}_{\text{ext}}(\omega)}{-i\omega M(\omega)} \quad (10)$$

has a pole wherever $M(\omega) = 0$. Inverting the Fourier transform,

$$\mathbf{v}(t) \propto \int d\omega \mathbf{v}(\omega) e^{-i\omega t} \quad (11)$$

a pole at $\omega = i/\tau$ in the upper half-plane contributes a term $e^{-i\omega t} = e^{t/\tau}$ that grows exponentially in time, which represents the unphysical runaway solution of the Abraham-Lorentz equation (16.9). A zero in the upper half- ω -plane thus signals a violation of causality and stability.

(c) To find the zeros for $\omega M(\omega) = 0$, we require

$$\begin{aligned} 0 &= \omega M(\omega) = \omega m + \frac{2e^2\omega}{3ac^2} \left(\frac{e^{i\xi} - 1 - i\xi}{2i\omega a/c} \right) \\ &= m \left[\omega + \frac{e^2}{3ia^2mc} (e^{i\xi} - 1 - i\xi) \right] = m \left[\omega - \frac{i}{2}\tau \left(\frac{c^2}{a^2} \right) (e^{i\xi} - 1 - i\xi) \right] \end{aligned} \quad (12)$$

The governing equation is thus

$$e^{i\xi} - 1 - i\xi = -\frac{2i\omega}{\tau} \left(\frac{a^2}{c^2} \right) = -i\xi \left(\frac{a}{c\tau} \right) \implies e^{i\xi} - 1 - i\xi \left(1 - \frac{a}{c\tau} \right) = 0 \quad (13)$$

With $\xi = x + iy$, (13) represents two simultaneous equations

$$\text{real part} \quad e^{-y} \cos x - 1 + y \left(1 - \frac{a}{c\tau} \right) = 0 \quad (14)$$

$$\text{imaginary part} \quad e^{-y} \sin x - x \left(1 - \frac{a}{c\tau} \right) = 0 \quad (15)$$

Obviously $\xi = 0$ is a trivial zero of (13). Let $\beta = 1 - a/(c\tau)$, our claim is that

$$\beta \leq 0 \quad \text{or} \quad a \geq c\tau \quad (16)$$

is the *necessary and sufficient* condition for the absence of a runaway zero ξ in the upper half-plane $y > 0$.

- **Sufficiency:** With $\beta \leq 0$, for any $\xi = x + iy, y > 0$, we have $|e^{i\xi}| = e^{-y} < 1$. But on the other hand $|1 + i\xi\beta| = |1 - y\beta + ix\beta| = \sqrt{(1 - y\beta)^2 + x^2\beta^2} \geq 1$, so (13) cannot be satisfied.
- **Necessity:** When $\beta > 0$, we can always find a nontrivial zero in the upper half-plane. Setting $x = 0$ automatically satisfies (15). Zeros of (14) are intersections of e^{-y} and $1 - y\beta$. They intersect at $y = 0$ (trivial solution). But at $y = 0$, e^{-y} has slope -1 , $1 - y\beta$ has slope $-\beta = -1 + a/(c\tau)$ – a slower negative slope than -1 . This means there must be another intersection for some $y > 0$ since the slope of $e^{-y} \rightarrow 0$ as $y \rightarrow \infty$. Therefore it is necessary to have $\beta \leq 0$ to avoid a runaway zero in the upper half-plane.

Note that the condition $a \geq c\tau$ is equivalently expressed as

$$m_0 = m - \frac{2e^2}{3ac^2} \geq m - \frac{2e^2}{3\tau c^3} = 0 \quad (17)$$

Zeros may still lie in the lower half-plane, which give a decaying solution, not violating causality and stability.

2. Prob 16.5

(a) For external field $\mathbf{E}_{\text{ext}}(t) = \mathbf{E}_0 \Theta(t)$, the Fourier transform of the external force is

$$\mathbf{F}_{\text{ext}}(\omega) = \frac{\mathbf{E}_0 e}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Theta(t) e^{i\omega t} dt = \frac{\mathbf{E}_0 e}{\sqrt{2\pi}} \int_0^{\infty} e^{i\omega t} dt = \frac{i\mathbf{E}_0 e}{\sqrt{2\pi}\omega} \quad (18)$$

where we have used $\omega \rightarrow \omega + i0^+$ for convergence.

By (10),

$$\mathbf{v}(\omega) = -\frac{\mathbf{E}_0 e}{\sqrt{2\pi}} \frac{1}{\omega^2 M(\omega)} \quad (19)$$

giving the time-domain velocity

$$\begin{aligned} \mathbf{v}(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathbf{v}(\omega) e^{-i\omega t} d\omega = -\frac{\mathbf{E}_0 e}{2\pi} \int_{-\infty}^{\infty} \frac{e^{-i\omega t}}{\omega^2 M(\omega)} d\omega \quad \omega = \frac{c\xi}{2a} \\ &= -\frac{\mathbf{E}_0 e a}{\pi c} \int_{-\infty}^{\infty} d\xi \frac{e^{-iT\xi}}{\xi^2 M(\xi)} \quad T = \frac{ct}{2a} \end{aligned} \quad (20)$$

(b) We evaluate the integral by closing the contour with a large semicircle. Since

$$|e^{-iT\xi}| = e^{Ty} \quad (21)$$

for the semicircle to vanish we need $Ty \rightarrow -\infty$, which means for $t \leq 0$, we must close in the upper half-plane. When condition (16) is satisfied, there are no zeros of $M(\xi)$ in the upper half-plane. For regularization $\omega \rightarrow \omega + i0^+$ to work in (18), we see the double pole given by ξ^2 in the denominator must lie in the lower half-plane, i.e., $\omega = -i0^+$, thus overall there is no poles in the upper half-plane. By Cauchy residue theorem, the upper half-plane contour integral vanishes, i.e., no preacceleration.